

A person standing in the air

Abu Dhabi University, August 2026

An image in your space cannot exceed the aperture

Every dimension in this design is a consequence of that one sentence, and none of it is a technology limit anyone can out-build.

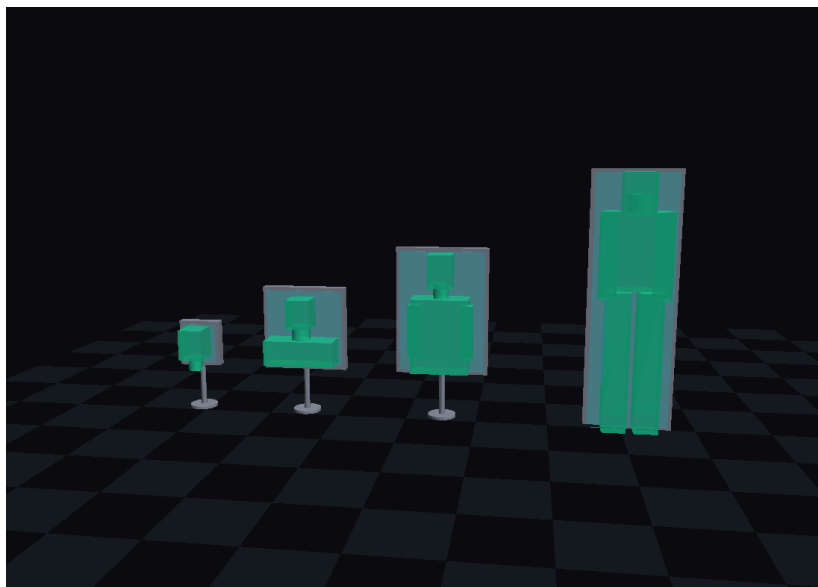
4× linear, 16× in area

the size of the error, not the plate

This project told its own author that a 20 cm plate shows a full upper body floating in open air. It shows 20 cm.

The correction was written the same night. The front page kept the claim for another six days, because the document that was fixed was not the document people read first.

Quantity	Value
an upper body measures	80 cm
the plate's aperture	20 cm
the image it can give	20 cm



Each plate is exactly as tall as the figure standing in front of it. That is the whole law, drawn at true scale.
It is the picture to return to whenever a proposal sounds too good.

To see a point, light has to enter your eye travelling from that point. There are only two ways that happens. Either the light started at the point, or it passed through the point from something behind it.

The second case is what a window does, and it is why a screen can show you a room it is not in. It also fixes where you may stand: behind the subject, from your position, there has to be something emitting.

Now ask for every direction at once. Nothing can be behind the subject from every direction simultaneously, so the second case is unavailable and the light has to leave the subject's own position.

With optics the only way to arrange that is to put aperture at every azimuth, which is the next slide's arithmetic and the reason this is a room rather than a device on a desk.

$$N = \frac{2\pi z}{D}$$

N

engines needed to tile the full circle at the image

z

distance from the image out to the aperture ring, in metres

D

width of one engine's aperture, in metres

D/z

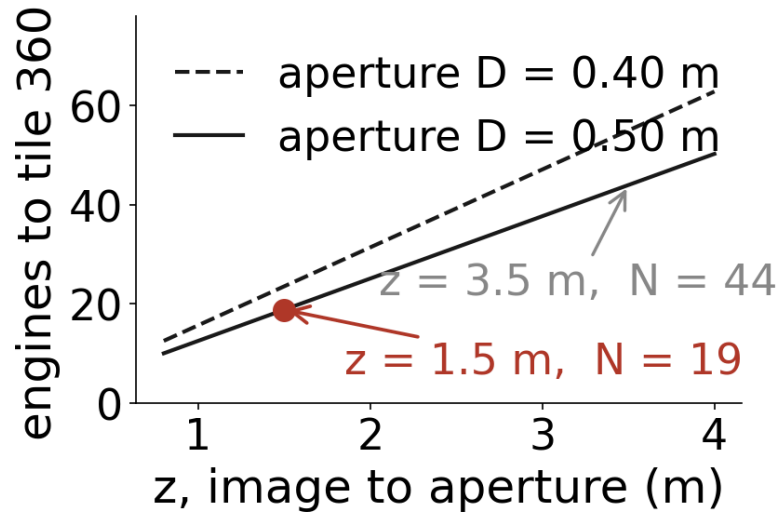
the angular wedge one aperture serves, in radians

$2\pi z$

the circumference every aperture together must cover

Each aperture serves a wedge about D/z wide, so tiling the circle takes $2\pi z/D$ of them. Angular wedges, viewer-ring patch width and étendue each returned the same count, which is the only reason I believed it.

Viewer distance does not appear anywhere in it, so what comes out is a room size rather than a seating plan.



The count scales with the radius, so the cheap room is the small one, and every instinct says otherwise.

Aperture ring radius	Engines at $D = 0.5$ m	Engine cost
1.2 m	15	\$38,925
1.5 m, the design point	19	\$49,305
2.5 m	31	\$80,445
3.5 m	44	\$114,180

Four people stand at a metre and a half from the centre of a four-metre room, each seeing the remote person from their own angle.

Quantity	Value
engines	19
relay band area	about 7 m ²
power	about 1.6 kW, one 20 A circuit
engine cost at the one quoted price	\$49,305

The engine count is set by geometry, so it is the one line of the budget that cannot be negotiated down, only designed down.



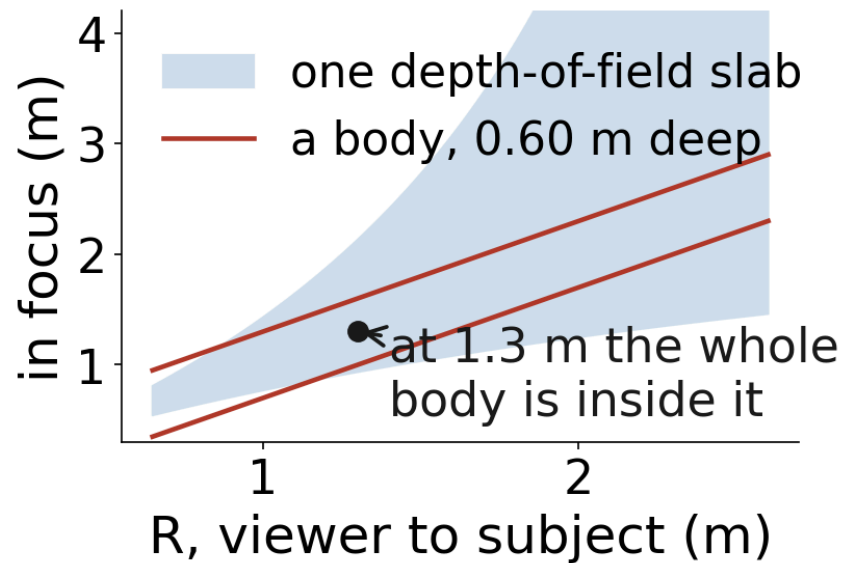
Viewers stand inside the aperture ring, so the ring radius has to exceed the viewing distance.

The first version had them at 1.3 m with the ring at 1.2 m, which puts the audience outside their own walls.

$R = 1.3$ m, $z = 1.5$ m, $N = 19$.

The eye does not resolve depth in millimetres

Converting distances into diopters deleted the most expensive component in this design.



The shaded band is everything held in focus at once. At one metre it is 66 cm thick; at a metre and a half, 1.7 m. The red lines are the front and back of a standing person, and inside the design range they sit entirely within it.

$$\Delta = \frac{1}{R - t/2} - \frac{1}{R + t/2}$$

Δ	t	R
the depth the eye actually sees, in diopters	the subject's front-to-back depth, in metres	distance from viewer to subject, in metres
t/R^2	0.60 D	what falls out
the first-order form, within 4% over this range	one depth of field, the threshold it is judged against	resolution dropping as the square of distance

A head spans half a diopter at 0.7 m and four hundredths of one at 2.5 m. Same head, same 25 cm of skull, a factor of thirteen in what the eye is given to work with.

Depth resolution falls as the square of distance, which is why intuition about it is unreliable: everyone calibrated theirs at arm's length, and a room is not arm's length.

Every number in this idea is that one conversion applied to a different question.



Dividing space gave 32 planes. Dividing diopters gives one, and the difference was a third of the budget.

Deleted by that one correction

swept-focus element

plane-switch rate

deformable mirror

tunable-lens investigation

risk register entry

What it had been

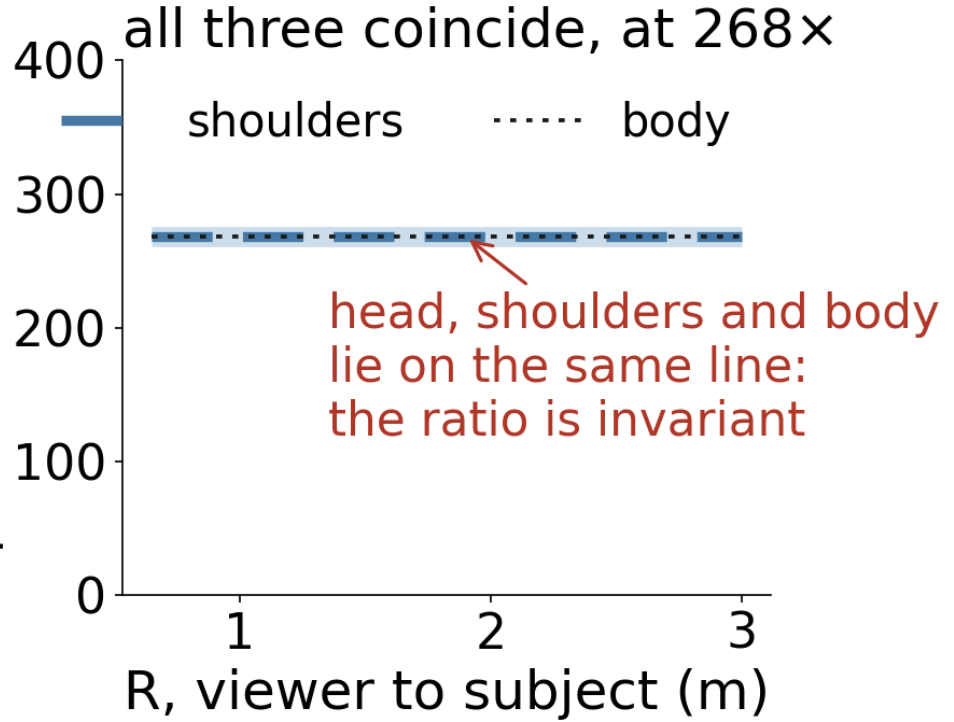
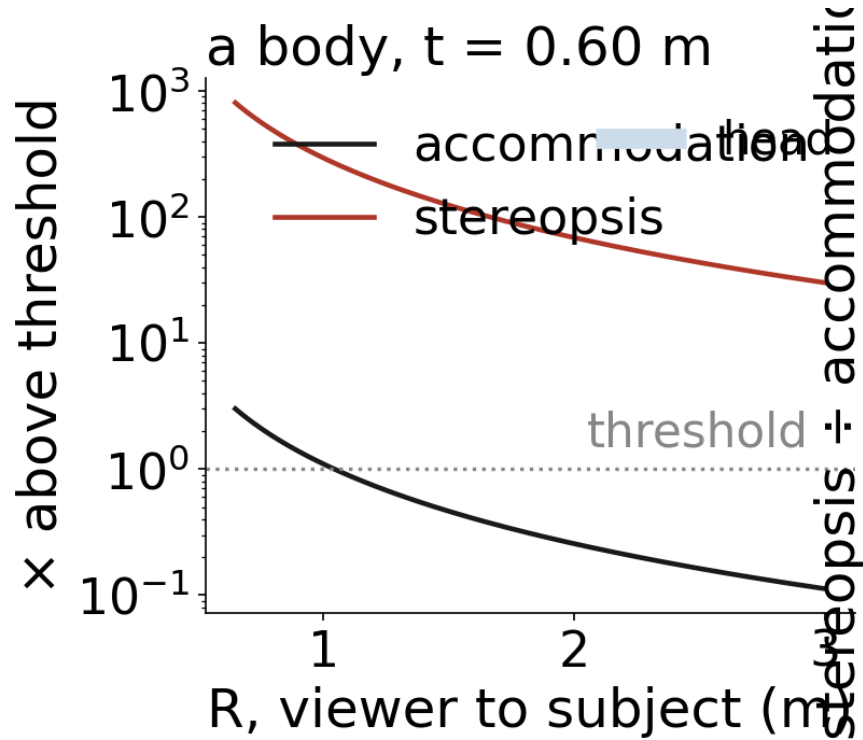
the highest-risk component

2,700 Hz

10kto50k each

two weeks of reading

ranked first of seven



Accommodation falls below threshold at about a metre. Stereopsis never comes near its own.

268×

stereopsis over accommodation, at every distance

Interocular separation times two depth-of-fields, over the stereo threshold. Subject size and viewing distance both cancel out, so this is a number and not a curve.

A display that gets disparity right and focus wrong is two orders of magnitude nearer correct than one that does the reverse. That is why headsets work at all.

Stereo threshold assumed	Ratio	Still beats focus?
10 arcsec, trained observers	804×	yes
30 arcsec, the working value	268×	yes
60 arcsec, clinical normal	134×	yes

A design review in July. The swept-focus element is ranked first of seven risks, and the plan is to spend six weeks and a borrowed deformable mirror on it.

Quantity	Before the conversion	After
focal planes	24 to 32	1
plane-switch rate	2,700 Hz	none needed
focus component	10kto50k each	deleted
engines needing one	19	0
risk register position	first of seven	removed

Converting a distance into a diopter took the most expensive and highest-risk item out of the design entirely. Not cheaper: absent.

What the conflict actually costs

Fatigue over a long session. It is measurable, it is the reason headsets are uncomfortable after an hour, and it is a real engineering target.

For a telepresence room, where a call runs an hour, comfort is worth paying for on its own terms.

What it does not cost

Depth accuracy. Accommodation contributes almost nothing to depth judgement at conversational distance, and slide 15 is the factor.

So a free-space image is not obviously better at showing depth than a well-made stereo screen. Its advantage has to be argued somewhere else, and that argument turned out to be about how many people are in the room.

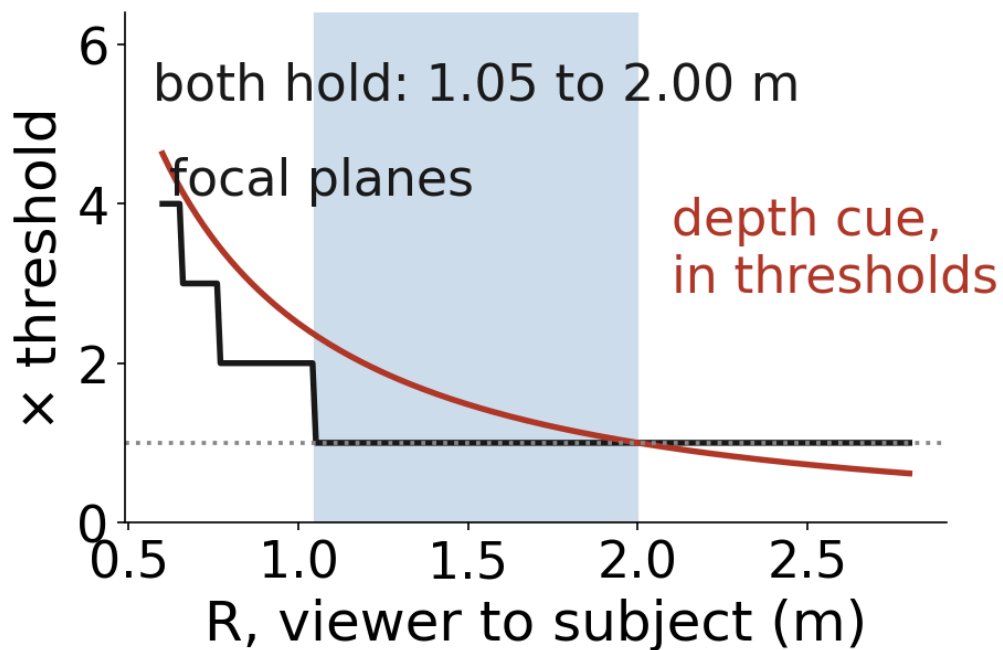
Lead the pitch with this, not with 360° .

docs/13_THE_ROOM.md §6 2026-08-21

<https://github.com/tamim1089/tayf>

The thing being led with was correct accommodation, which slide 15 shows is the weakest depth cue the eye has.

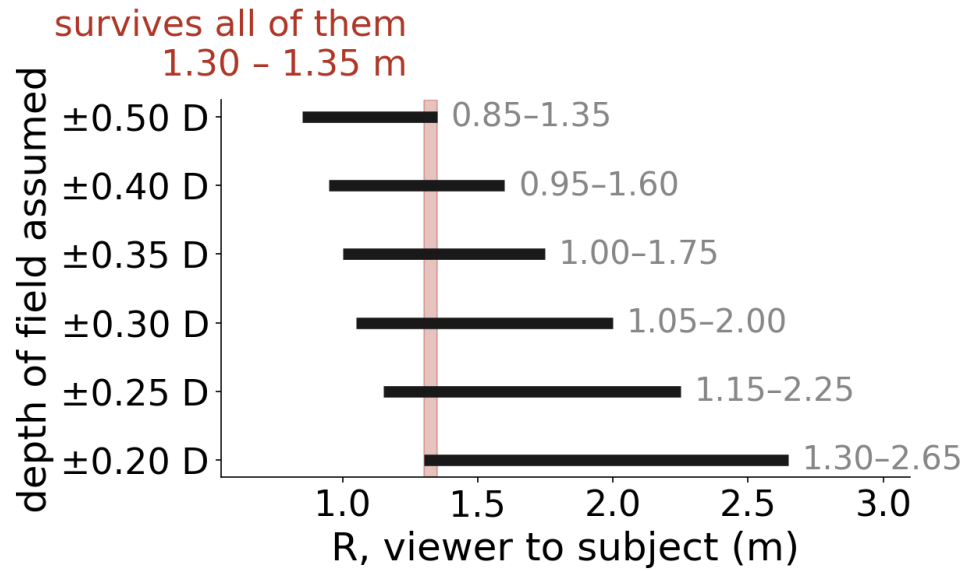
The correction sits four sections below it in the same file, because the rule here is to fix the error where it happened rather than quietly delete it.



Too close and the body spans several diopters and needs more than one focal plane. Too far and the cue against the background falls below threshold.

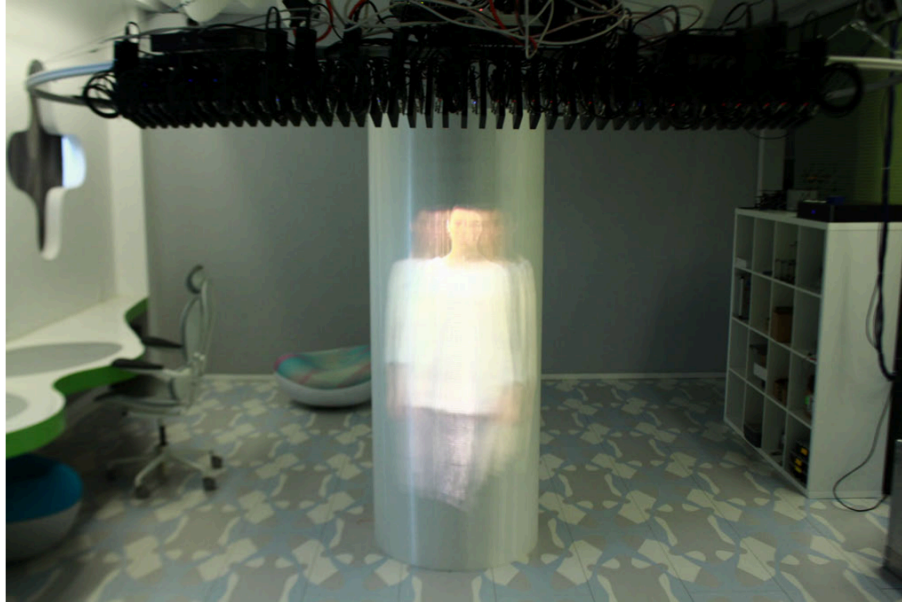
Below threshold the free-space advantage is not subtle, it is absent.

Both hold only between about 1.05 m and 2.00 m.

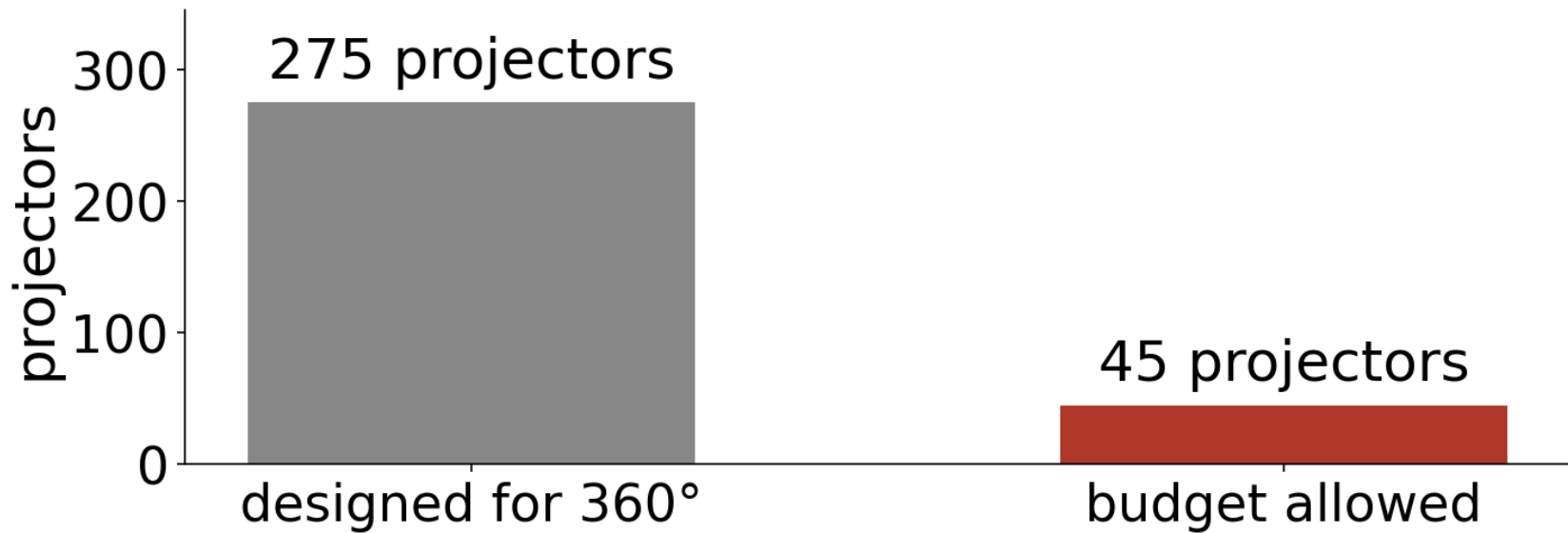


The depth-of-field number is the one assumption everything rests on, and I have it from a secondary reading. So rather than defend it, sweep it, and take the viewer distances that work for every value in the range.

A ring of projectors around a cylinder, eight years ago



The architecture derived in the last two ideas, photographed in 2018.



The rail was machined for a full circle at one projector per 1.3° , which is the angle the retroreflective sheeting returns light within. The budget covered one sixth of it.

Forty-five times 1.3 is fifty-nine degrees, and fifty-nine degrees is exactly what the system delivered.

The system as delivered	Value
projectors needed for 360°	275, at one per 1.3°
projectors installed	45
parallax delivered	59°
cylinder	195 cm tall, 75 cm diameter, acrylic
surface	retroreflective sheeting, 1.3° return
vertical parallax	none, traded for a diffuser
render	about 10 frames per second
capture	ZED stereo cameras at 15 fps

The coating is a surface, so accommodation is fixed at 75 cm of acrylic no matter what the picture shows.

I searched the inventor's patent estate expecting to find the blocking patent.

Patent family	Subject
five members, 2003	attentive interfaces
two members, 2003	calibration-free eye tracking
three members, 2006	flexible displays
two members, 2021	gaze estimation, for Huawei
cylindrical light-field telepresence	none

Checked four ways: inventor enumeration, and the university as assignee crossed with each of cylindrical display, telepresence and light field. All zero.

Broadcast: every view, always, to everyone

275 engines, because the retroreflective sheeting returns light within 1.3° of where it came from, so every viewing direction needs its own projector.

Almost all of that light lands where no eye is. The system pays for every viewpoint whether or not anyone occupies it, which is what makes the count a property of the room rather than of the audience.

Steered: light only where eyes are

19 engines for the same coverage, which is where the étendue argument said it would land before I had heard of the 2018 system.

The cost is a tracker, and the tracker has to fail dark. A tracker that fails bright is a safety incident rather than a glitch, which is why the eye-safety case and the cost case are the same argument.

The same four-person meeting as slide 9, served by the 2018 system and by this one.

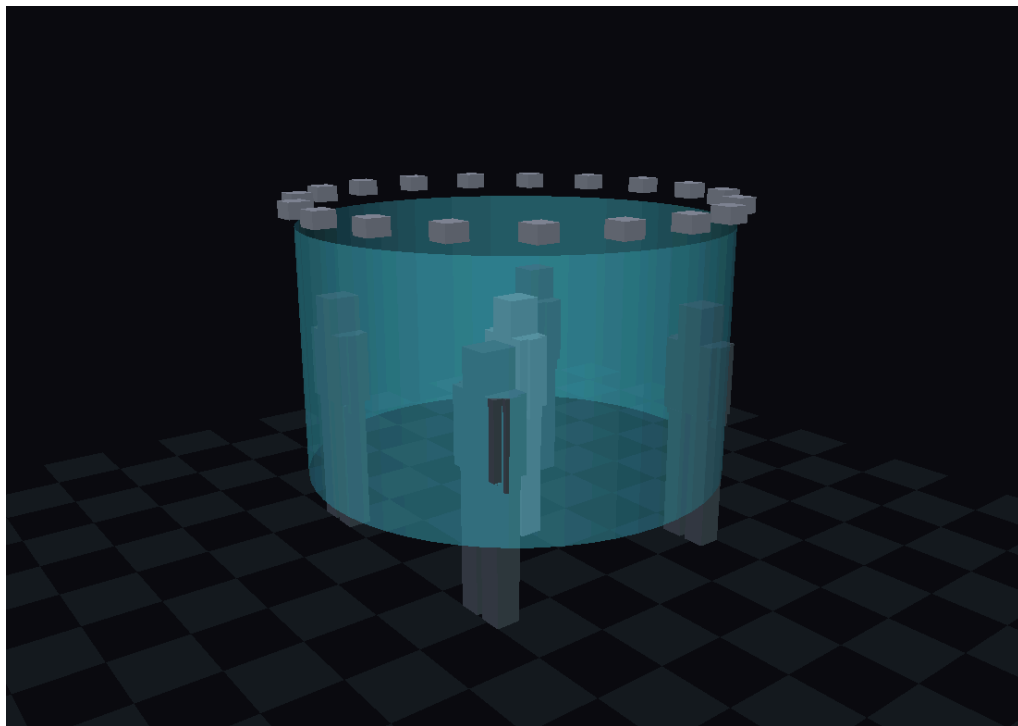
	TeleHuman 2, 2018	This design
coverage	59°	360°
frame rate	about 10 FPS	video rate
where the image sits	on an acrylic tube	in the air
vertical parallax	none, traded away	±20°
engines	45 of 275 needed	19

Serving several viewers at once is the one claim a tracked stereo screen cannot match, because such a screen serves exactly one viewer by construction.

Three constraints, in the order they bind.

Quantity	Value	Fixed by
R , viewer to subject	about 1.3 m	the window on slide 19
z , image to aperture	1.5 m	$z > R$: they stand inside
D , one aperture	0.5 m	what an engine can throw
N , engines	19	$N = 2\pi z/D$, slide 6
cost of another 10 cm	2.3 engines	the same law

No step in that chain is a preference, which is what to say when someone asks whether the room could be bigger.



The corrected geometry: nineteen engines on the ring, the relay band, and the audience inside it.

The version before this one had the audience standing outside the walls of the room meant to contain them.

A test now asserts $z > R$ in the model source.

Element	Position	Ours to own?
the ring architecture	published 2018, unpatented	no, and nobody's
pupil steering	anticipated, active to 2040	no
the avatar pipeline	anticipated, published	no
the relay band	no prior art found	yes
multi-engine calibration	no prior art found	yes, narrowed

What survives is the band and the calibration, which is a thin moat, and saying so now is cheaper than hearing it in a diligence meeting.

The differentiator might not exist

If a person fits inside one slab of focus, a free-space image may not be distinguishable from a screen at the same place.

5.2 against 5.3

co-presence, flat cutout against full 3D avatar

A published study put a life-size, correctly placed flat video cutout against a full three-dimensional avatar and found co-presence statistically indistinguishable.

It beat the avatar on rated fidelity. I found this in my own repository's notes, filed months earlier under a heading I had not read.

Measure	Flat cutout	3D avatar
co-presence, out of 7	5.2	5.3
fidelity, out of 7	5.1	3.7

Display	Disparity	Parallax	Multi-viewer	Opaque
a real object	yes	yes	yes	yes
the aerial image	yes	yes	yes	no
flat screen, same place	no	no	yes	yes
screen on the far wall	no	no	yes	yes
tracked stereo screen	yes	yes	no	yes

The experiment's conditions are read off this table rather than chosen. Against a real object the aerial image differs in exactly one column, and that column is why it is the calibration condition.

Can people tell?

The detection cells: a one-sided test against chance.

A significant result means the difference is perceptible, which is what most of the study asks.

Are these the same?

The calibration cell wants a null, and a non-significant result is absence of evidence, not evidence of absence.

Reporting it as indistinguishable would be a lie told with correct arithmetic, so it gets an equivalence test with the margin fixed before any data exists.

Fault the simulation found**What it would have cost**

between-subject spread used as the total

five subjects too few

decision needed all three cells at once

joint power 0.51, not 0.80

rig gate confused two different claims

a good rig read as broken

three calibration tests uncorrected

15% false condemnation

six distances in one sitting

173 minutes per subject

Every one of these would have surfaced only as the word inconclusive, after fifteen subject-hours were spent.

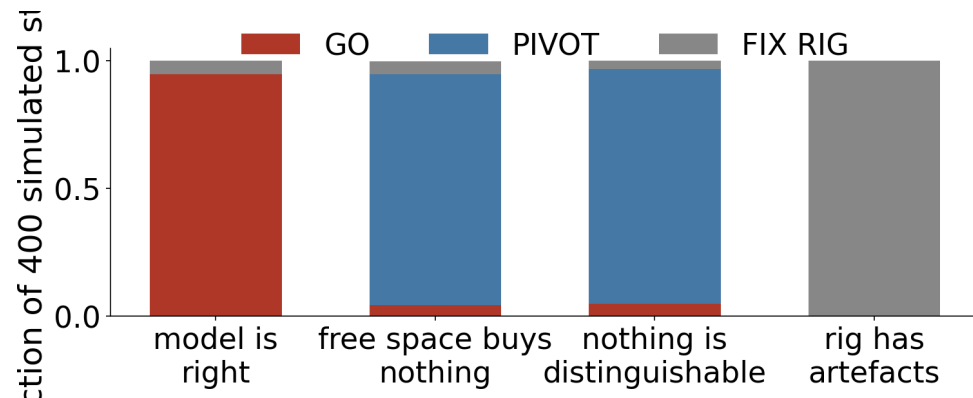
15%

of good rigs condemned by an uncorrected test

Three calibration tests, each at five percent, none of them corrected for the fact that there were three of them.

The bench cost is the obvious half. The power loss is the half nobody notices, because the study still runs and still reports a number.

Consequence	Effect
bench time	days rebuilding a working rig
study power	15% lost, and silently
power, before and after	0.85 to 0.95



Four worlds I constructed, four hundred simulated studies in each, and the decision the analysis returned.

Error rate under the null	Intended	Measured
primary cell, uncorrected by design	0.050	0.050
secondary family, corrected	≤ 0.050	0.040
calibration family, corrected	≤ 0.050	0.033
six cells left uncorrected		0.265

One person, one sitting, a bench in a dark room, and a stepper motor swapping what sits at the image point.

Quantity	Value
subjects	26
trials each	180
minutes each	36, one sitting
subject-hours	15.6
parts	\$215

That is the whole cost of finding out whether the idea is worth \$141,500 of hardware.

An analysis chosen after seeing the results is a search for a p-value, so this one is frozen first.

Outcome	What it means	What happens
free space beats a same-place screen	the claim holds	build the wedge
no difference from that screen	the claim fails	pivot, for \$215
rig fails calibration	we cannot tell yet	fix the optics first
chance everywhere	the cue model is wrong	the most useful outcome

Fixed number of subjects, no peeking, and no adding people after looking at the data. Those three sentences are what separate an experiment from a search.

The same bench closes a second number that no published source states.

Measurement	Status
retroreflector return efficiency	never published
what depends on it	every brightness figure here
cost to close it	one borrowed luminance meter
order	first, because it sets the screen dimming

The source has to be a physical object rather than a screen: the relay images whatever it is fed, so a flat source gives a flat aerial image and the experiment becomes a null by construction.

```
# three p-values, corrected for being three
p = {'a': 0.001, 'b': 0.030, 'c': 0.031}
holm(p, alpha=0.05)

# it sorts first, then steps down, and stops
# at the first threshold it misses. Nothing
# after that point can reject, however small.
# → {'a': True, 'b': False, 'c': False}

# my first worked example put a small p after
# a failure. Sorting makes that impossible.
```

p	threshold	outcome
0.001	0.0167	reject
0.030	0.0250	retain, stop
0.031	0.0500	retained
		under 0.05 and still fails

Both wrong examples are still in the test file as comments. A test that encodes a mistake is worse than no test.

One researched price broke the plan

The cost model was arithmetically consistent and its inputs had never been researched.

\$2,195

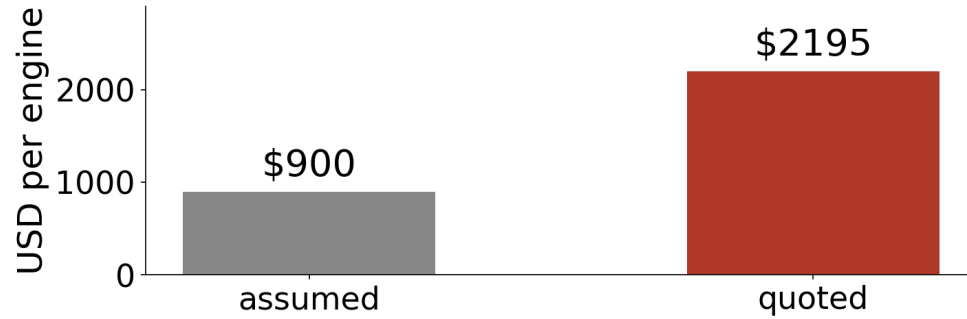
one board, one vendor page, one afternoon

Every other number in the bill of materials was engineering judgement. This one is a price a supplier publishes for a part in stock.

It replaced an assumed \$900, and the difference propagated further than a single line usually does.

	Value
assumed, per engine	\$900
quoted, per board	\$2,195
engines in the room	19

One vendor page moved the engine line by a factor of three, and took the margin with it.



Line	Assumed	Researched
per display engine	\$900	\$2,595
19 engines	\$17,100	\$49,305
prototype total	\$101,000	\$141,500
volume total	\$42,000	unresolved
boards alone at qty 1		\$41,705

At the price I can verify, delivered cost exceeds the list price I had been quoting.

Claim	Status
prototype cost	stands, \$141,500
volume cost	withdrawn, no quote exists
\$180,000 list price	withdrawn
70% gross margin	withdrawn
recurring revenue	stands, independent of the hardware

One written volume quote from either named supplier restores all four, and nothing else does.

What this needs

6.6 m² angularly multiplexed band

one bespoke part

a master at band scale

a pre-revenue customer

a quote

What the supplier has shown

finished film to 1400 mm wide

automotive parts to 400 × 300 mm

master printer at A2

automotive OEM design cycles

not yet requested

The relay band is the only part nobody has patented, which makes it the moat, and essentially one firm makes it, which makes it the single-source risk.

A call with one end in the European Union, carrying a live photoreal likeness of a real person.

Obligation	Detail
classification	a deepfake under Article 50
in force since	2 August 2026
marking grace period	to 2 December 2026
tier	limited risk, not high risk
fine ceiling for a small company	five figures, not €15M

The cryptographic session binding already needed against impersonation is the compliance mechanism. One control answers both.

The patent I warned about

Claim 1 is tied to an emissive array of angular pixels, and to a hierarchical tracking sequence that locates the viewer, then the head, then the eyes.

A ring of projectors throwing converging wavefronts onto a relay band plausibly practises neither, which is a strong literal non-infringement position and a weak reason to relax, because the doctrine of equivalents exists.

The estate I had not counted

About 391 active patents, several of them running past 2035, over converging wavefronts, energy-relay structures and plenoptic rendering.

That is this design's own vocabulary read back at it. It is where a genuinely blocking claim would come from, it needs a paid opinion rather than my reading, and I had not counted it at all until somebody else did.

Open question**What it costs to answer**

can anyone perceive the difference

\$215 and 15.6 subject-hours

retroreflector efficiency

one borrowed instrument

will the band supplier quote

one email

engine price at volume

one email

freedom to operate

a patent attorney's fee

Two of the five decide everything and cost almost nothing. That ordering is the only project-management insight here.

The next thing that happens costs \$215

Settled, and checked by 132 tests: the aperture decides the size, the tiling law decides the count, and the component that dominated the budget was never needed.

Not settled: whether anyone can perceive the difference. No further reasoning will settle it, which is why the next artefact is a bench.

A beamsplitter, two figurines and a stepper, for \$215.